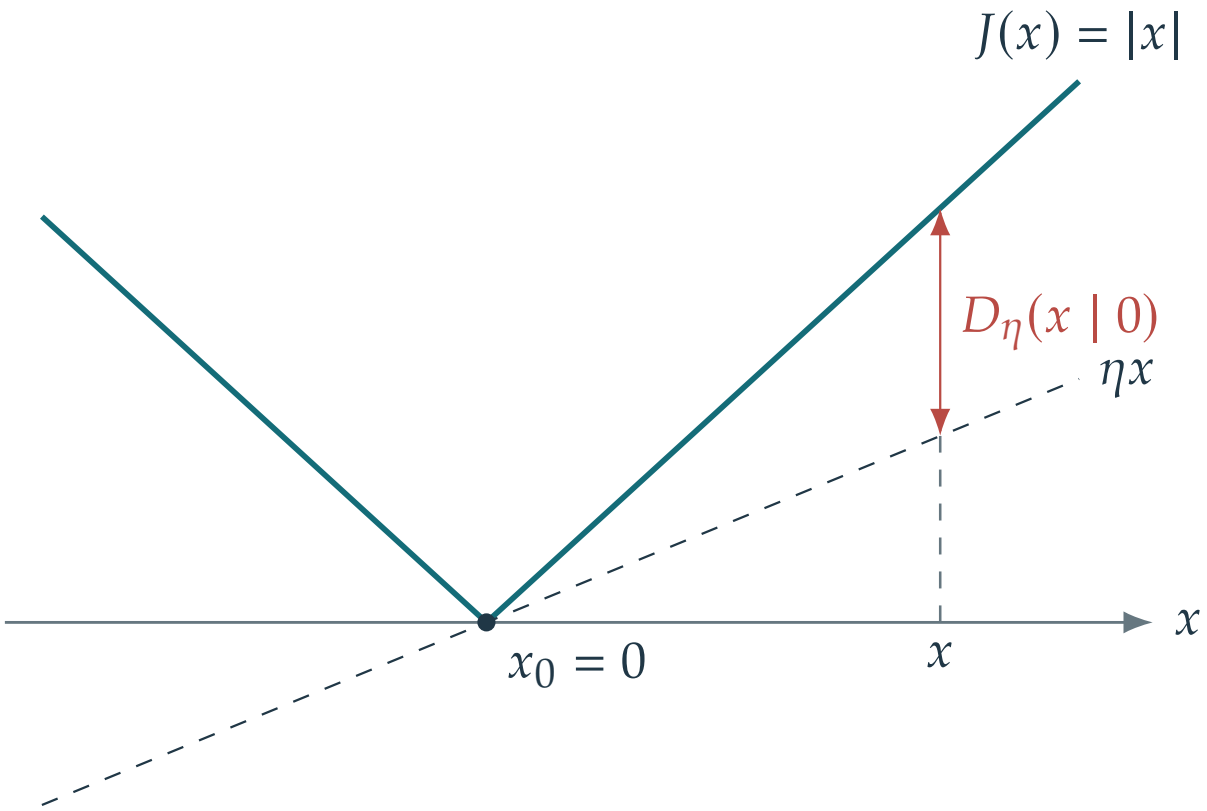

$$|\eta| < 1$$

$$D_\eta(x \mid 0) = |x| - \eta x$$

$$\geq (1 - |\eta|)|x|$$

The margin $1 - |\eta|$ is positive precisely when the subgradient does not saturate.